Concentration Inequalities

Markov, Chebyshev, and Chernoff

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Why Concentration?

The Question

A random variable X has some expectation $\mathbb{E}[X] = \mu$.

But a single draw of X is **random**: it need not equal μ.

Concentration
How likely is X to land **far** from its mean?

$$\Pr\big[\, |X - \mu| \geq t \,\big] \;\leq\; ?$$

Concentration inequalities give **tail bounds**: guarantees that X is close to μ with high probability.

Markov's Inequality

Markov's Inequality: Statement

Markov's Inequality
If $X \geq 0$ is a **non-negative** random variable, then for any $a > 0$,

$$\Pr[\,X \geq a\,] \;\leq\; \frac{\mathbb{E}[X]}{a}.$$

- The bound says: X cannot exceed many times its mean too often.
- Setting $a = k\,\mathbb{E}[X]$ gives $\Pr[\,X \geq k\,\mathbb{E}[X]\,] \leq \frac{1}{k}$.

Markov's Inequality: Proof

Start from the definition and keep only the large terms:

$$
\begin{aligned}
\mathbb{E}[X] &= \sum_x x \Pr[X = x] && \text{(definition of expectation)} \\
&\ge \sum_{x \ge a} x \Pr[X = x] && \text{(drop terms with } x < a\text{; all terms } \ge 0\text{)} \\
&\ge a \sum_{x \ge a} \Pr[X = x] && \text{(each remaining } x \ge a\text{)} \\
&= a \Pr[X \ge a].
\end{aligned}
$$

Dividing both sides by $a > 0$ gives $\Pr[X \ge a] \le \dfrac{\mathbb{E}[X]}{a}$. ■

Markov: Example

Suppose the average score on an exam is $\mu = 60$ (scores are ≥ 0).

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Suppose the average score on an exam is $\mu = 60$ (scores are ≥ 0).

What fraction of students could have scored at least 90?

$$\Pr[X \geq 90] \leq \frac{60}{90} = \frac{2}{3}.$$

At most $\frac{2}{3}$ of students scored ≥ 90.

Weak but assumption-light: we used *only* the mean — nothing about the spread of scores. Markov is often loose, but it is the foundation for the stronger bounds.

Variance

Variance: Definition

Markov uses only the mean. To get a sharper bound we need to measure how **spread out** X is around its mean $\mu = \mathbb{E}[X]$.

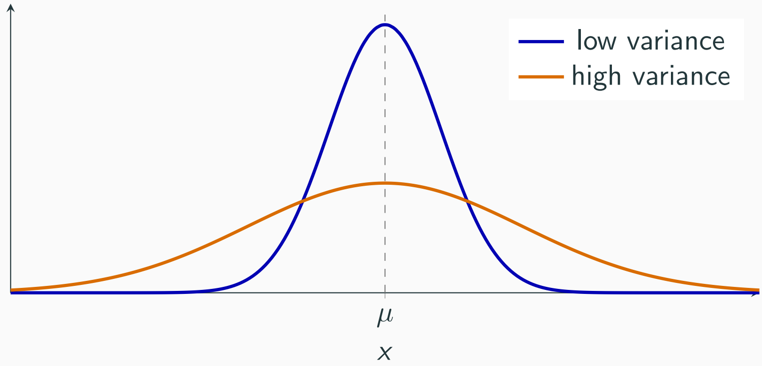
Variance

$$\mathrm{Var}[X] \;=\; \mathbb{E}\big[(X-\mu)^2\big] \;=\; \sum_x (x-\mu)^2 \Pr[X=x].$$

Variance is the **average squared distance** from the mean.

Variance: Picture

Both distributions have the **same mean** μ, but different spread.



Low variance: mass packed tightly around μ. **High variance:** mass spread far out — larger deviations are common.

Variance: The Computational Formula

Expanding the square gives a formula that is usually easier to compute:

$$\begin{aligned}\mathrm{Var}[X] &= \mathbb{E}\big[(X-\mu)^2\big] = \mathbb{E}\big[X^2 - 2\mu X + \mu^2\big] \\ &= \mathbb{E}[X^2] - 2\mu\,\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2.\end{aligned}$$

$$\mathrm{Var}[X] \;=\; \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

Recipe: compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2] = \sum_x x^2 \Pr[X = x]$, then subtract the square of the mean.

Variance: Scaling and Shifting

Scaling and shifting

$$\mathrm{Var}[aX + b] \;=\; a^2\,\mathrm{Var}[X].$$

- **Shifting by b:** moving every value by a constant slides the whole distribution — it does not change the spread, so b drops out.
- **Scaling by a:** stretches every deviation by a, and deviations are *squared*, so the variance scales by a^2.
- Consequence: the standard deviation scales by $|a|$, $\sigma[aX + b] = |a|\,\sigma[X]$.

Variance: Sum of Independent Variables

Sum of independent variables
If $X_1, \ldots, X_n$ are independent, then

$$\mathrm{Var}\left[\sum_i X_i\right] \;=\; \sum_i \mathrm{Var}[X_i].$$

- Independence is required here.
- Contrast with expectation: $\mathbb{E}[\sum_i X_i] = \sum_i \mathbb{E}[X_i]$ holds *always*.

Variance: Why Averaging Helps

Let $X_1, \ldots, X_n$ be independent, each with mean μ and variance σ^2, and let $\bar{X}_n = \frac{1}{n} \sum_i X_i$.

Combining the two properties:

$$\mathrm{Var}[\bar{X}_n] = \mathrm{Var}\left[\frac{1}{n} \sum_i X_i\right] = \frac{1}{n^2} \sum_i \mathrm{Var}[X_i] = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

- The mean stays $\mathbb{E}[\bar{X}_n] = \mu$, but the variance **shrinks like** $1/n$.
- More samples $\Rightarrow$ the average concentrates around μ.
- This is exactly the fact Chebyshev will turn into a tail bound next.

Chebyshev’s Inequality

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For any random variable X with mean μ and variance $\sigma^2 = \mathrm{Var}[X]$, and any $t > 0$,

$$\Pr\big[\,|X - \mu| \geq t\,\big] \;\leq\; \frac{\sigma^2}{t^2}.$$

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Proof. Apply Markov to the non-negative variable $Y = (X-\mu)^2$:

$$\Pr\big[\,|X-\mu| \geq t\,\big] = \Pr\big[\,(X-\mu)^2 \geq t^2\,\big] \;\leq\; \frac{\mathbb{E}[(X-\mu)^2]}{t^2} = \frac{\sigma^2}{t^2}.$$

Now we use the **variance** — so a tightly-spread X gets a better bound.

Chebyshev: The Law of Large Numbers

Recall for an average of n i.i.d. samples, $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathrm{Var}[\bar{X}_n] = \sigma^2/n$.

Plugging $\bar{X}_n$ into Chebyshev:

$$\Pr\big[\,|\bar{X}_n - \mu| \geq \varepsilon\,\big] \;\leq\; \frac{\mathrm{Var}[\bar{X}_n]}{\varepsilon^2} \;=\; \frac{\sigma^2}{n\,\varepsilon^2}.$$

- For *any* fixed accuracy ε, the bound $\to 0$ as $n \to \infty$.
- The sample average converges (in probability) to the true mean.

This is a one-line proof of the **(weak) law of large numbers**.

Chebyshev: Example

X has mean $\mu = 3.5$ and variance $\sigma^2 = 3.25$.

Bound the chance of deviating by 3 or more:

$$\Pr\big[\,|X - 3.5| \geq 3\,\big] \;\leq\; \frac{3.25}{3^2} \;=\; \frac{3.25}{9} \;\approx\; 0.36.$$

How many independent samples to estimate μ within $\varepsilon = 0.1$ with failure probability ≤ 0.05?

$$\frac{\sigma^2}{n\,\varepsilon^2} \leq 0.05 \;\;\Longrightarrow\;\; n \;\geq\; \frac{3.25}{0.05 \cdot (0.1)^2} \;=\; 6500.$$

Chernoff Bounds

Chernoff: Sums of Independent Indicators

Let $X = \sum_{i=1}^{n} X_i$ where the $X_i \in \{0,1\}$ are **independent**, and let $\mu = \mathbb{E}[X]$.

Chernoff Bound (multiplicative form)
For any $0 < \delta < 1$,

$$\Pr[\,X \geq (1+\delta)\mu\,] \leq e^{-\mu\delta^2/3}, \qquad \Pr[\,X \leq (1-\delta)\mu\,] \leq e^{-\mu\delta^2/2}.$$

Combined: $\Pr\big[\,|X-\mu| \geq \delta\mu\,\big] \;\leq\; 2\,e^{-\mu\delta^2/3}.$

The tail decays **exponentially** in μ — vastly stronger than the $1/t^2$ from Chebyshev.

Chernoff: Example (Fair Coins)

Flip a fair coin $n = 1000$ times; let X be the number of heads, $\mu = 500$.

Chance of at least 600 heads, i.e. $\delta = 0.2$:

$$\Pr[\, X \geq 600 \,] \;\leq\; e^{-\mu\delta^2/3} \;=\; e^{-500 \cdot 0.04/3} \;\approx\; e^{-6.7} \;\approx\; 0.0012.$$

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Compare **Chebyshev** ($\sigma^2 = np(1-p) = 250$, $t = 100$):

$$\Pr[\,|X - 500| \geq 100\,] \;\leq\; \frac{250}{100^2} = 0.025.$$

Chernoff: ~ 0.001 vs. Chebyshev: 0.025 — far tighter.